ROOFING ESTIMATE

COX ROOFING & RESTORATION — XACTIMATE FORMAT

STANDARD + MOST MISSED

Generated: 3/19/2026 at 1:54:08 PM

HOMEOWNER **(GIS server unavailable)**	OWNER MAILING ADDRESS **7325 PRINCETON AVENUE, ST. LOUIS, Missouri**
PROPERTY ADDRESS **7325 PRINCETON AVENUE, ST. LOUIS, Missouri**	PARCEL ID **MANUAL_38.659075_-90.325989**

ROOF SYSTEM **Ludowici Spanish Barrel Tile (13-1/4")**	TOTAL AREA **2,614 SF (26.1 SQ)**	WITH WASTE (20%) **31.3 SQ**
PITCH **6/12**	STORIES **1**	LINE ITEMS **19**
MEASUREMENT SOURCE **LiDAR Geometry (T-shape, 85% confidence)**		

TOTAL ESTIMATE — STANDARD + MOST MISSED

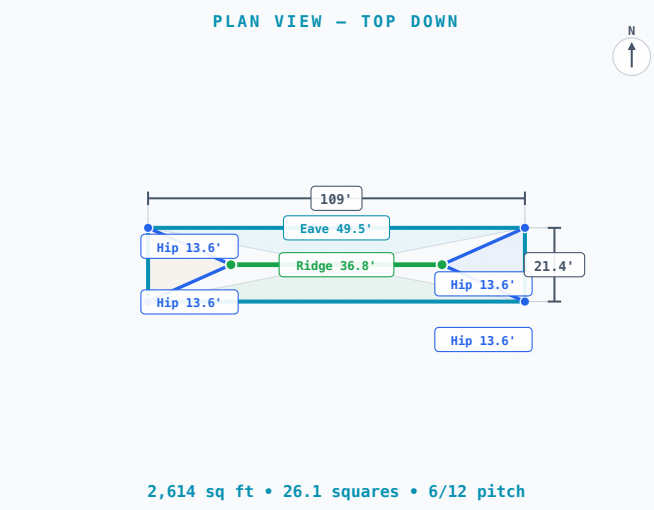
$111,603.01

19 Xactimate line items

ROOF DIAGRAMS & MEASUREMENTS

7325 PRINCETON AVENUE, ST. LOUIS, Missouri — Cox Roofing & Restoration

PLAN VIEW — TOP DOWN



3D PERSPECTIVE VIEW

MEASUREMENT SUMMARY

TOTAL ROOF AREA **2,614 SF**	ROOF SQUARES **26.1 SQ**	PITCH **6/12**	STORIES **1**
RIDGE LF **37 LF**	HIP LF **54 LF**	VALLEY LF **49 LF**	EAVE LF **118 LF**
RAKE LF **62 LF**	DRIP EDGE LF **181 LF**	PERIMETER **390 LF**	WASTE FACTOR **20%**

ROOF SYSTEM — LUDOWICI SPANISH BARREL TILE (13-1/4")

LUDOWICI • Measurement source: LiDAR Geometry (T-shape, 85% confidence)

Roof diagrams are schematic representations based on GIS building footprint and AI measurements. Not to scale.

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LINE ITEMS

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CODE	DESCRIPTION	QTY	UNIT	UNIT PRICE	TOTAL
RFG TILE LUDOWICI	R&R Ludowici Spanish Barrel Tile (13-1/4")	31.3	SQ	$2439.80	$76365.74
RFG TEAR TILE	Remove tile roofing	26.1	SQ	$374.26	$9768.19
RFG IWB HT	Ice & water barrier - high temp (for tile)	3137	SF	$1.83	$5740.71
RFG RDG LUDOWICI	R&R Ludowici hip roll (#102)	91.2	LF	$62.67	$5715.50
RFG DRPE COPPER	R&R Drip edge - copper	180.6	LF	$9.77	$1764.46
FLS PJ	Flashing - pipe jack	6	EA	$55.94	$335.64
GUT COPPER 5	R&R Gutter - copper - 5"	118.3	LF	$45.00	$5323.50
GUT ALU 5 DR	Gutter/downspout - Detach & reset — required on all roofing scopes	118.3	LF	$4.52	$534.72
RFG STORAGE	Material storage / staging charge — required on all roofing scopes	1	EA	$150.00	$150.00
FLS STEP COPPER	R&R Step flashing - copper	20	LF	$21.48	$429.60
FLS PJ	Flashing - pipe jack	6	EA	$55.94	$335.64
RFG TEAR2 LAM	2nd Layer Tear-Off — Additional layer removal — missed when adjuster doesn't verify layer count	26.1	SQ	$61.94	$1616.63
SHT OSB 12	OSB Sheathing Replacement — Decking replacement — missed when adjuster doesn't inspect under shingles. Estimate 10% of area.	261	SF	$2.89	$754.29
FLS CHIM AVG	Chimney Flashing (Average) — Chimney flashing replacement — frequently missed or underscoped	1	EA	$422.67	$422.67
GUT ALU 5	Gutters (Full Replacement) — Gutter replacement — often missed when gutters are damaged during tear-off	118.3	LF	$9.64	$1140.41
GUT GUARD	Gutter Guards — Gutter guard replacement — missed when existing guards are removed during tear-off	118.3	LF	$4.42	$522.89
RFG VENT TRTL MTL	Turtle Vents (Replace) — Static vent replacement — missed when vents are damaged during tear-off. Estimate 3.	3	EA	$75.90	$227.70
RFG VENT PWR	Power Vent (Replace) — Power attic vent — missed when existing unit is damaged. Estimate 1.	1	EA	$270.78	$270.78
RFG CORN LAM	Gable Cornice Return — Cornice return at gable ends — frequently missed. Estimate 2.	2	EA	$91.97	$183.94
				TOTAL	$111603.01

MOSL8X Price List (Feb 2026) — Sales-grade estimate based on AI measurements and GIS building footprint data.

This is not a binding contract. Quantities derived from STL City GIS building footprint polygons and AI roof analysis. Actual quantities may vary upon physical inspection.

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WEATHER DATA & STORM HISTORY

7325 PRINCETON AVENUE, ST. LOUIS, Missouri — Cox Roofing & Restoration

HAIL IMPACT ANALYSIS

DATE	SIZE	MAGNITUDE	LOCATION	DISTANCE	STATUTE STATUS
2025-05-15	**Golf Ball / Hen Egg**	2"	University City	1.19 mi	**57 DAYS LEFT**

MISSOURI STATUTE OF LIMITATIONS

Property damage claims must be filed within **1 year** of the storm event date per Missouri Revised Statutes §516.120. Hail events within 3 miles of the property are documented above.

WIND DESIGN SPEED
115 mph (ASCE 7-22, Risk Category II)

CLIMATE ZONE
4A (Mixed-Humid)

EXPOSURE CATEGORY
Exposure B (urban/suburban)

AVG JANUARY TEMP
30°F (triggers ice barrier requirement)

ADJUSTER NOTE — WEATHER & WIND

- St. Louis is in ASCE 7 wind speed zone requiring 115 mph basic wind speed for Risk Category II structures.
- Roof coverings must be rated for the design wind speed per IBC 1504.1.
- Hip roofs receive a 10% reduction in wind uplift pressures compared to gable roofs (ASCE 7 Figure 30.3-2A).
- Edge and corner zones experience 2-3x higher wind pressures than field of roof — enhanced fastening required.
- Climate Zone 4A requires ice barrier at eaves per IRC R905.1.2 (avg Jan temp ≤25°F threshold is met).
- Minimum R-30 continuous insulation above deck for flat roofs per IECC R402.2.1.
- Vapor retarder required on warm-in-winter side of insulation per IECC R702.7.1.
- Freeze-thaw cycles average 80-100 per year — affects material selection and expansion joints.

Weather data sourced from NOAA Storm Events Database, NWS Local Storm Reports, and ASCE 7-22.

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BUILDING CODES & PERMITS

7325 PRINCETON AVENUE, ST. LOUIS, Missouri — Cox Roofing & Restoration

JURISDICTION	BUILDING CODE	PERMIT COST
St. Louis City / St. Louis County, MO	IRC 2021 / IBC 2021	$150.00 - $350.00

IRC/IBC CODE REFERENCES — STEEP ROOF

CODE	REQUIREMENT
IRC R905.1.2	Ice barrier required at eaves in areas where avg daily temp in January is 25°F or less — applies to St. Louis climate zone
IRC R905.2.7	Underlayment for asphalt shingles — one layer of #15 felt or ASTM D6757 synthetic over entire roof
IRC R905.2.7.1	Ice barrier — self-adhering polymer modified bituminous sheet from eave edge to ≥24" past interior wall line
IRC R905.2.8.5	Drip edge required at eaves and gables — 2" min lap, corrosion-resistant metal, proper fastening
IRC R905.2.6	Asphalt shingle attachment — 4 nails per shingle min, 6 nails in high-wind areas (≥110 mph)
IRC R905.2.4.1	Slope requirements — asphalt shingles min 2:12 slope (4:12 recommended), special underlayment for 2:12-4:12
IRC R905.2.5	Asphalt shingle product standards — comply with ASTM D3462 or D7158 Class G/H
IRC R806.1	Ventilation — minimum 1 sq ft NFA per 150 sq ft attic area; 1:300 with balanced intake/exhaust
IBC 1507.2	Weather protection — roof assemblies shall be designed for weather protection per manufacturer installation
IBC 1504.1	Wind resistance — roof coverings shall be designed for wind loads per ASCE 7

LOCAL AMENDMENTS

- St. Louis County Code of Ordinances Chapter 1115; Ice barrier requirement R905.1.2 applies for climate zone
- St. Louis City local amendments and IRC 2021 require ice barrier on all roof surfaces to a point at least 24 inches inside the exterior wall line

No building permits found for this property in the STL City ArcGIS database.

ASTM MATERIAL STANDARDS — STEEP ROOF

The following ASTM standards govern material specifications for this steep roof system.

ASTM CODE	STANDARD TITLE	DESCRIPTION
ASTM D3462	**Asphalt Shingles Made from Glass Felt**	Specifies physical properties, dimensions, and test methods for fiberglass-based asphalt shingles.
ASTM D7158	**Wind Resistance of Sealed Asphalt Shingles (Uplift Force)**	Classification for wind resistance — Class D (≤90 mph), G (≤120 mph), H (≤150 mph). St. Louis requires Class G minimum.
ASTM D3161	**Wind Resistance of Asphalt Shingles (Fan-Induced)**	Fan-induced method for wind resistance classification — Class A (60 mph), Class F (110 mph).
ASTM D1970	**Self-Adhering Polymer Modified Bituminous Sheet Materials**	Ice and water shield requirements — mandatory at eaves, valleys, and penetrations per IRC R905.1.2.
ASTM D6757	**Inorganic Underlayment for Steep Slope Roofing**	Synthetic underlayment performance — tear resistance, UV exposure, and slip resistance.
ASTM D4869	**Asphalt-Saturated Organic Felt Underlayment**	Standard for #15 and #30 felt underlayment used beneath asphalt shingles.
ASTM D228	**Asphalt Roll Roofing (Organic Felt)**	Specifications for mineral-surfaced and smooth roll roofing products.

Building codes enforced by St. Louis City Building Division. Permit data from STL City ArcGIS. ASTM standards per IRC/IBC requirements.

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PERMEABILITY & CODE COMPLIANCE

7325 PRINCETON AVENUE, ST. LOUIS, Missouri — Cox Roofing & Restoration

PERMEABILITY & VAPOR BARRIER REQUIREMENTS

Vapor permeability requirements for Climate Zone 4A (St. Louis). Proper vapor management prevents condensation, mold, and premature roof assembly failure.

CODE	TITLE	REQUIREMENT
IRC R702.7	Water-Resistive Barrier	Exterior walls require a water-resistive barrier behind cladding — must have permeance ≥5 perms (Class III) or ≥1 perm (Class II) per ASTM E96.
IRC R806.5	Unvented Attic Assemblies	Unvented attics require air-impermeable insulation at the roof deck — vapor retarder ≤1 perm on warm side per IECC Table R702.7.1.
ASTM E96	Water Vapor Transmission of Materials	Standard test method for measuring water vapor permeance (perms) of building materials — used to classify vapor retarders (Class I: ≤0.1 perm, Class II: 0.1-1.0 perm, Class III: 1.0-10 perms).
IECC R702.7.1	Vapor Retarder Requirements	Climate Zone 4A (St. Louis): Class I or II vapor retarder required on warm-in-winter side of insulation. Exception: vented attic assemblies.
IRC R905.2.7	Underlayment Permeability	Asphalt shingle underlayment must allow adequate vapor transmission — synthetic underlayments must meet ASTM D6757 permeance requirements.

ADJUSTER NOTE — CODE COMPLIANCE & SUPPLEMENTS

The IRC/IBC building codes, ASTM material standards, and permeability requirements listed above are **mandatory** for all permitted roofing work in St. Louis City/County. When a roof is replaced, it must be brought up to current code — this includes ice barrier, drip edge, proper ventilation, and vapor retarder requirements that may not have existed when the original roof was installed. These code-required upgrades are **supplementable items** that should be included in the insurance claim scope.

ASTM standards referenced per IRC/IBC requirements. Permeability data per IECC Climate Zone 4A.

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